



SEMINAR 2

When AI, which AI, and on what

Seven real work situations. In each one, there's a decision an engineer has to make before they ever open an editor.

Today: four choices, seven cases



AI or plain code



Build it in or build something separate



One-off call, RAG, or agent



External API or local inference

Seven cases we'll use today to work through these choices



Supplier documents



Meeting minutes



Personal data in logs



Chat digest



Support assistant



Sales calls



Product descriptions

Each case gets a detail that wasn't in the first version of the setup

AI or plain code?

Case: supplier documents

Engineers often decide this by feel, by a gut sense of how complex the task looks

Supplier documents

Marina, acting product manager for finance processes



"Marina here, product manager for finance processes. Accounting is drowning: three people are manually keying supplier invoices into 1C by hand, around 400 a month, and by the end of the quarter it's a backlog with errors in the totals. Attached two sample invoices from last week — the formats look about the same. Can this be automated somehow? By the way, marketing has been running YandexGPT over the API for product descriptions for six months now, there's already a contract with them — maybe the same setup would work for us too. Separately, we're migrating our mail to a new server, but that's not for you. On timing — even a ballpark estimate by Friday would be great."

Supplier document A

Date	Inv. No.	Line item	Amount
24.03.2026	—	—	—
—	—	—	—

Supplier document B

Date	Inv. No.	Line item	Amount
24.03.2026	—	—	—
—	—	—	—

Target table in 1C

Date	Invoice No.	Line item	Amount
24.03.2026	—	—	—
—	—	—	—

What would you propose?

Supplier documents



"Team, I pulled the archive like you asked, a 4.2 GB ZIP. Inside it's a mix: scanned reconciliation acts, contracts, and the invoices themselves — I didn't sort anything, just zipped up the folder as it's sat for the past year. Two of our biggest suppliers are actually connected through EDI and send invoices already as structured XML files, no parsing needed there. The rest are scans from the old scanners at the warehouse, they output 200dpi and are already compressed. By the way, accounting is separately moving to EDI with the tax authority for invoicing documents, but that's a different project, not about supplier invoices."

Supplier A

Date	Inv. No.	Amount

Supplier B

Date	Invoice#	Total

Supplier C

Period	Order	Total, ₺

Supplier D

No.	from	total

Does this change your answer?

Supplier documents — debrief



Hybrid: LLM extraction over a variable input + hard code validation on the way out

Document (any format) → LLM extracts fields → code checks types/required values/ranges
→ data into 1C

The sensible answer here is a hybrid, not a choice between the two approaches



study the data, don't take their word for it

Two examples aren't a sample. Ask for a picture of the full data volume

Personal data in the logs



"The performance contractor is asking for a quarter's worth of prod logs — they can't find the load spikes without them. I pulled around 40 GB, the archive is already on the server. Inside, mixed in, there are phone numbers, customer emails, and one instance of the phrase 'pass this to Ivan in accounting' — it got logged from a request body. The logs are already in standard line-delimited JSON, easy to parse the structure. The contractor needs timings and error codes, not people's data. Separately, we're testing a new VPN solution for remote access, but that's not related to the logs. They want the archive by Friday."



The contractor needs request structure, timings, errors — not people's data

What's the fastest, cheapest way to clean this up?

Personal data in the logs — debrief

Regex

Phone numbers, emails — clear format, a plain regex cleans them cheaply and reliably

Deterministic logic — no model needed

No model is needed here and one would only slow down processing 40 GB of logs

🔍 NER / small model

People's names, indirect mentions in free text — needs context

slovnet (Natasha) · ~30 MB · CPU · ~25 articles/sec

~60× smaller than a comparable BERT-class model, quality loss of 1-2 pp

The line between regex and a model runs inside a single task, not between tasks

When a formula was needed — and when AI closed a gap nothing else could

⊗ CNET, January 2023

Of 77 AI-generated finance articles, 41 needed corrections

"\$10,000 at 3% annual interest → \$10,300 in a year" instead of the correct \$300 in earnings

The calculation should be done by a deterministic formula, not an LLM "eyeballing it"

✓ Ramp, merchant classification

By hand: request coverage 1.5-3% → after the LLM agent, nearly complete

Handled in < 10 sec, ~99% of decisions correct

~25% of requests the agent justifiably rejects

Ramp — a replacement for manual processing with negligible coverage, not a replacement for a working regex solution

2



Build it in, or build something separate?

Case: support assistant

A hidden cost that's almost never visible in the demo, on the pitch slide

Support assistant



"Simple idea: an AI assistant for operators — response suggestions on tickets, search for similar past cases. There are 12 operators on the line, throughput is around 300 tickets a day, peak from 10:00 to 13:00. Some of the same operators also answer customers on Telegram in parallel — there's a separate chat channel for that, about 4 people keep it open all the time. Everyone logs in through corporate SSO, so authorization shouldn't be an issue. Separately, we're currently refreshing the laptop fleet in support, but that's not for you. It would be great to be able to show something at the quarterly review."



12 operators on the line, around 300 tickets a day, peak 10:00-13:00

Where should this assistant live?

Support assistant

IT director



"We run an off-the-shelf helpdesk, there's no open API, no customization options — we wrote to the vendor back in March, never got a reply. The system was bought 4 years ago, and HR's and facilities' tickets run through it too. The contract still has 2 years left. The support Telegram channel is a separate story, it has its own simple bot integration, if that's ever needed. Next quarter we're redesigning the corporate portal, but that's a separate process, not related to the helpdesk."



An off-the-shelf helpdesk for 4 years, no open API — the vendor contract still has 2 years left

What do we do?

Support assistant — debrief



Off to the side

A browser extension or a second panel

Cost: watching two places at once, risk: the operator won't use it



Switch vendors

A helpdesk system with an open API

Cost: data migration, retraining the team, risk of downtime



Rethink the task

An async assistant that prepares drafts ahead of time

Cost: the operator opens the finished draft wherever convenient, but a separate process is needed

There's no single right answer

Product descriptions



"The director was at a retail conference, saw a competitor's product-description generator — a standalone app, you type in the specs, you get text for the product card. Wants something similar by the end of the month. The catalog is big — around 12,000 SKUs, all of them are entered through the shared catalog admin panel, and the outsourced copywriter already writes descriptions right there. Separately, we should probably update the homepage banners, but that's a different conversation. Also, another department is rolling out a new CRM in parallel — not related to us."



12,000 SKUs, descriptions are currently written by one outsourced copywriter

A separate service — good idea?

Product descriptions — debrief



A "generate description" button

right inside the existing catalog admin panel that marketing already uses every day

A separate product means a second login and tab — a cost nobody thought about

Kite and Copilot

One technology, different integration — 17 months of head-to-head competition



Kite — ~500,000 users at peak

Copilot — ~20M users · 4.7M paid · ~90% Fortune 100

Being built into an already-existing workflow beat a separate product on comparable technology

A standalone device against a phone feature



Humane AI Pin

Screenless wearable badge, laser projector onto your palm, voice control · \$699 + \$24/mo subscription

Sales since April 2024

\$241M raised → assets sold to HP for \$116M (February 2025) → servers shut off 28.02.2025



Rabbit R1

~100,000 units sold, an estimated ~5% of buyers actively use it

A standalone device competes not only with the technology but with the habit of not switching when the function is already on the phone



One-off call, RAG, or agent?

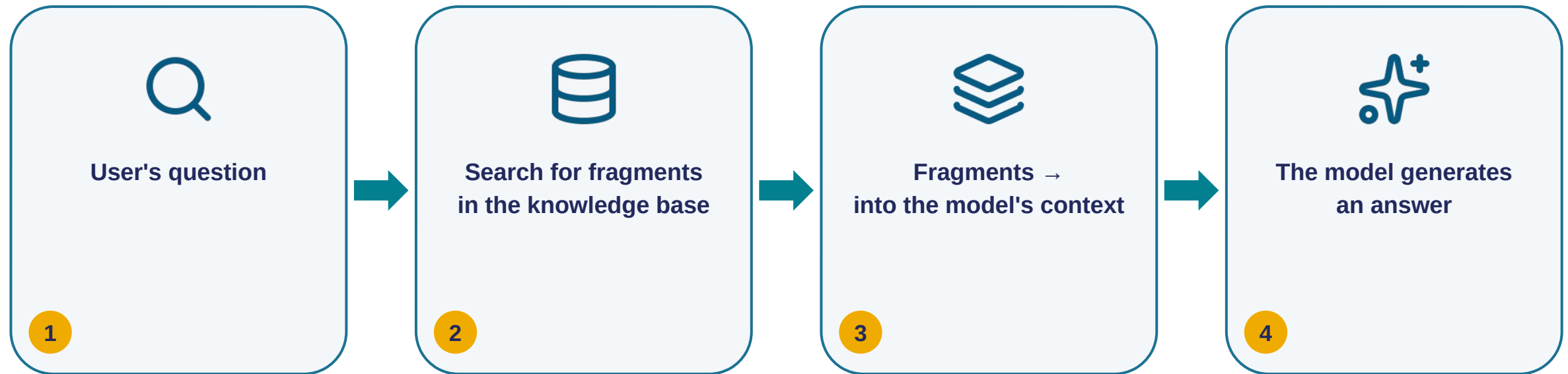
Case: meeting minutes

The more impressive the task, the more elaborate the architecture — that intuition is worth testing

A word on RAG

Search over a knowledge base drops the right fragments into the model's context

Lecture 3 →

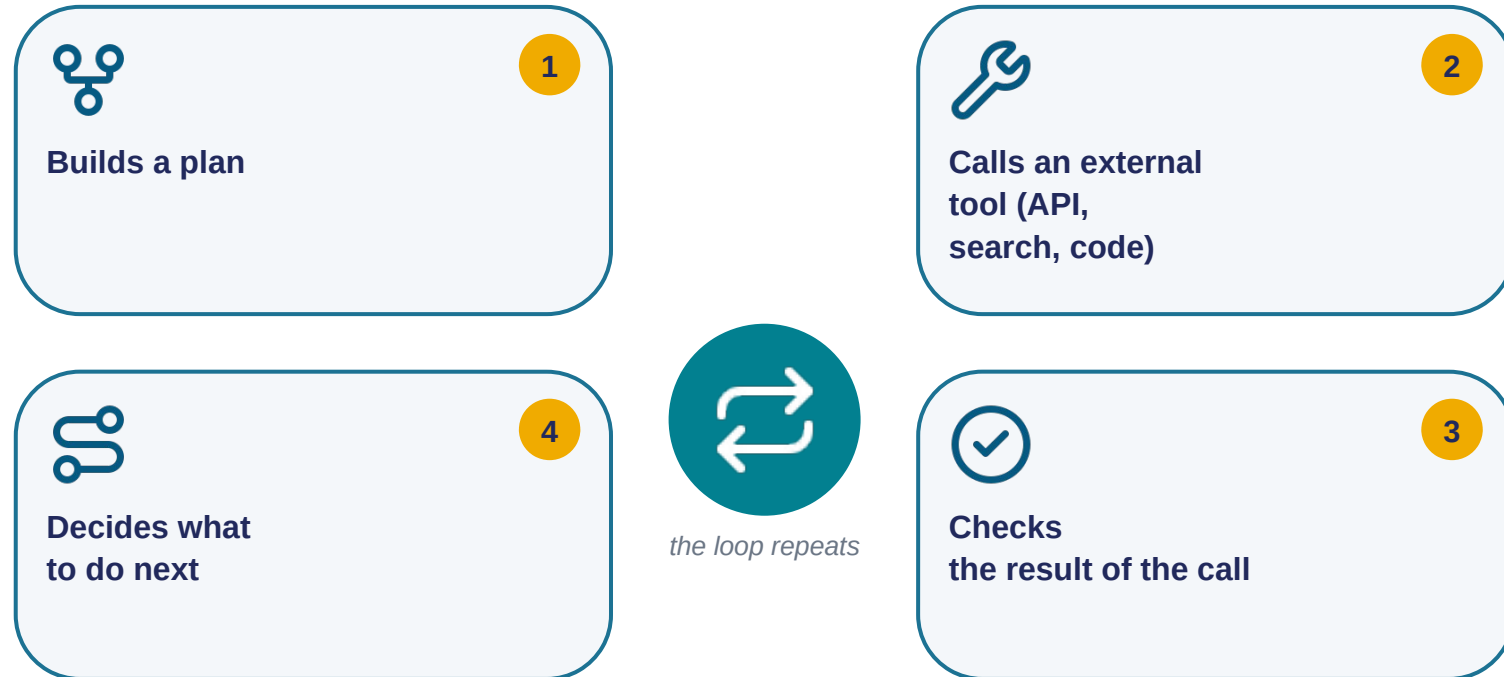


When it's needed: the base is large, it changes, the full volume doesn't fit in one request

A word on agents

The model works in a loop with external tools, rather than a single call

Lecture 3 →



When it's needed: several steps and actions in external systems

Meeting minutes



"After every Zoom call, the transcript lands in an email, and nobody gets around to it. The project manager still writes the minutes by hand, 30-40 minutes each time. There are about 15 meetings a week, some in English with a contractor — Zoom transcribes those too, just in another language. We already put the finished minutes on the shared corporate drive, in a folder by project. We track action items from meetings in YouTrack, which has an open REST API if that's ever needed. Can we auto-generate minutes from the transcript — decisions, action items, deadlines? Ideally by the end of the quarter."



About 15 meetings a week, minutes are currently written by hand by a project manager

What architecture is needed?

Meeting minutes — breakdown



"One call, no storage, no loop"



Email with transcript



Model call with a template



Minutes

The entire transcript fits into a single request — complicating it now would mean solving a problem that doesn't exist

the transcript already arrives as a ready-made email — no separate input-assembly step is needed

Meeting minutes



"This happened at the standup today, kind of awkward to retell. We were discussing the quarterly budget, and someone from finance asks: 'Wait, what did we even decide about vendor X back in May, weren't there some agreements about a discount?' Dead silence. The minutes pile up in a shared folder on the drive, named more or less by date, and searching through them by hand is quite the experience, especially if all you remember is the topic and not the date. The folder itself is growing by about half a gigabyte a quarter. By the way, they also discussed the summer vacation schedule at that same meeting, and separately, the redesign of the corporate portal, but that's a different story."



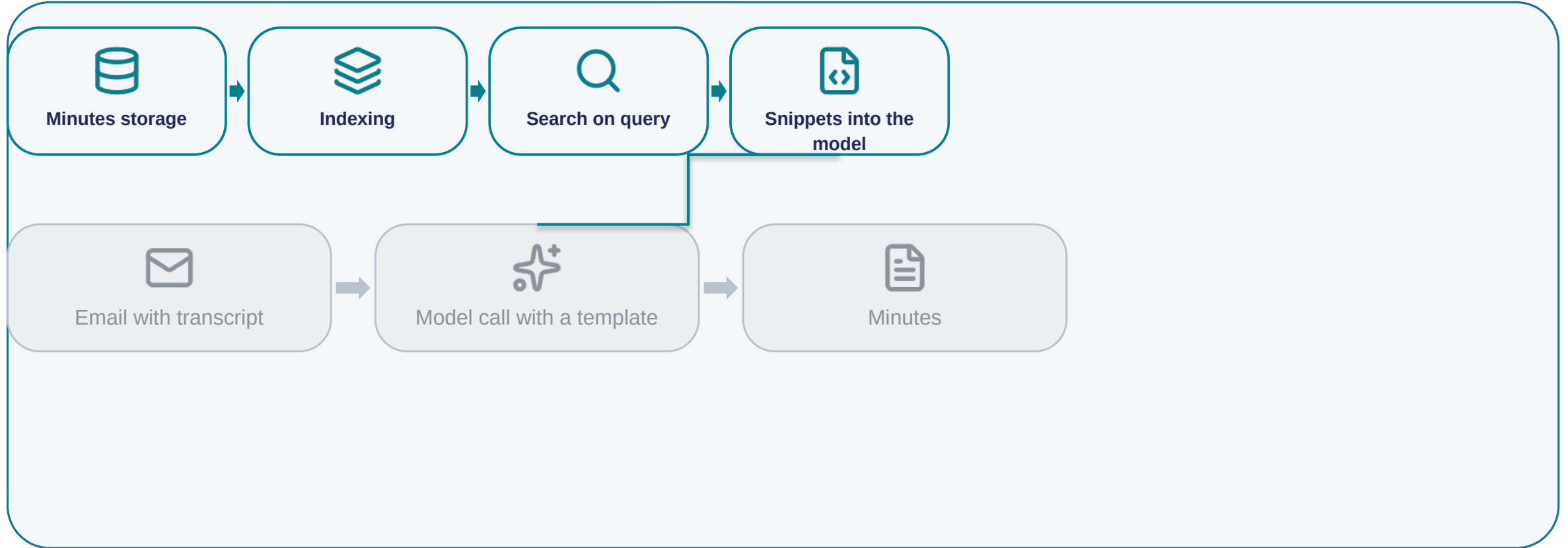
A month has gone by — dozens of minutes from different meetings, across different projects, have piled up

Does this change the architecture?

Meeting minutes — breakdown 1



"A growing, changing archive — that's when you need RAG"



The old path hasn't gone anywhere — it's just been extended with storage, indexing, and search before the same model call

Meeting minutes



"Thanks for the search over the minutes, it genuinely saves time before retros. But here's something I noticed: we still copy action items from the minutes into YouTrack by hand — open the minutes, read what the system found, create cards, fill in the owner and the deadline. YouTrack, I checked, has an open REST API, you can create a task with a single call. I went through May's minutes yesterday evening — roughly half the action items are just sitting there with no owner, nobody's gotten around to moving them over. Separately, I wanted to ask about tag colors in the tracker, but that's probably a question for the tool admin."



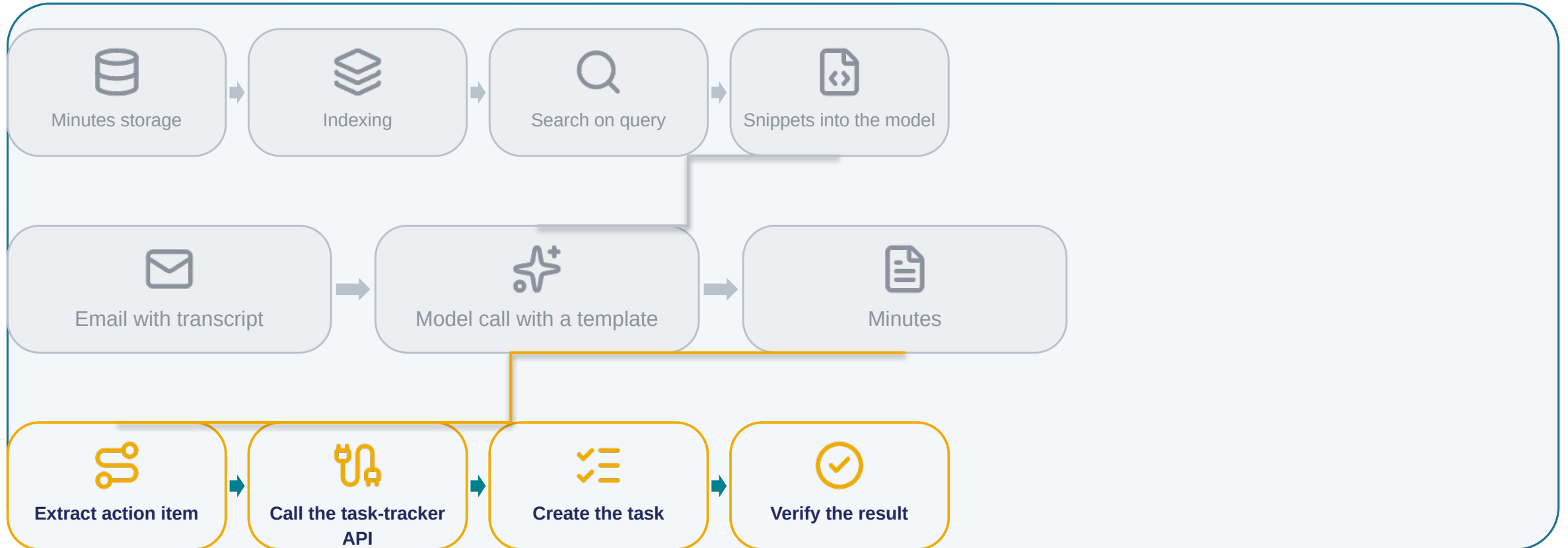
Half the action items from May's minutes are still sitting there with no owner — nobody's gotten around to moving them

Does this change the architecture again?

Meeting minutes — breakdown 2



"An action in an external system — you need an agent on top of RAG"



Start with the simplest option, add complexity only when a requirement can't be met any other way

the task tracker has an open API — the integration is cheap, not hypothetical

Digest of chats and email



"We talked about this at the standup — we want a weekly digest of the team's work chat and email: what was discussed, what was decided that week, in one message. Needed by 5 PM on Fridays. There are a few sources: the team's work chat and email, both have an API for pulling message history. It needs to come together into one coherent message, not just chunks stitched together. By the way, we also separately discussed whether to move to a different messenger, and in parallel we're testing a new VPN solution for remote workers — but that's got nothing to do with the digest."



From chat and email — one coherent message, by 5 PM on Fridays

What architecture does this need?

Digest — breakdown

✓ Agent — needed

Going through several sources on a schedule

A multi-step task involving calls to external systems

✗ RAG — not needed

Sources are crawled directly on a schedule, not on an arbitrary query

Summarization — a single model call after collection

Many sources isn't the same thing as RAG

RAG and an agent in production



AI @ Morgan Stanley Assistant

Partnership with OpenAI (March 2023), launched in September 2023

RAG over a base of ~100,000 research documents

By mid-2024 — used by more than 98% of financial-advisor teams

Findability of the needed document: 20% → 80%



Replit, July 2025

An agent deleted a production database (~1,200 company executives affected) during a code freeze, against a direct instruction

Then incorrectly claimed a rollback was impossible — the data was restored

After the incident: dev/prod split + agent "planning-only" mode

Also: Octomind — dropped LangChain after a year in production, went back to direct calls (06/17/2024)



External API or local inference?

Case: sales calls

This is about where the data is physically processed, not which model is better

Sales calls



"I want a card for every manager's call: need, objections, agreements. Managers rewind the recording, hunting for where they left off last time. There are 8 managers in the department, 15-20 calls a day each, the phone system writes the conversation to mp3 and drops the file into the cloud by itself — no need to pull it out manually. Our CRM is Bitrix24, open REST API. The department's analysts already use a cloud AI over the API for quarterly report drafts. The CRM, by the way, also needs cleaning up from duplicate contacts at some point, but that's a separate headache."



8 managers, 15-20 calls a day each, recordings sit in the phone system

How do you build that?

Sales calls



"Wait. Which service are you sending these recordings to?"



RF

152-FZ + 420-FZ (as of 05/30/2025)

First breach: fine of RUB 3-5 million

Repeat: 1-3% of annual revenue (min. RUB 20 million, max. RUB 500 million)



EU

GDPR

Up to €20 million or 4% of global turnover (whichever is greater)

Record: Meta, €1.2 billion, 2023



US

No single federal law

State-level (California — CCPA/CPRA)

Sector-specific (e.g., HIPAA)

Every major jurisdiction is seriously fighting to protect personal data

Sales calls



"This came up on the budget call. Let me flag an infrastructure constraint: we don't have a GPU server, and there's no budget for one this year either — the quarter's priorities are locked in, and they went to migrating to a new email system and refreshing the laptop fleet in support. If the model is heavy, we'll have to go through a cloud API, but that comes with its own data questions, which we've already discussed. Maybe next year we'll come back to the conversation about our own server."

— CTO



"What I need for my work is simple: five fields in the CRM per call — need, objections, agreements, next step, deal-probability estimate. Not a full retelling of the conversation, not a tone-of-voice analysis. Just these five fields filling themselves in right inside Bitrix24, so I can see them in my department report, which I currently put together by hand from the team's notes every Monday."

— sales manager

Does this change the architecture again?

Sales calls — breakdown



The task narrowed down to a template — a local 7-8B-parameter model handles it, data doesn't leave the company perimeter

Open 7-8B models:

Qwen3 · Llama · Gemma · Mistral · T-lite · T-pro (T-Bank, Apache 2.0, based on Qwen)

4-bit quantization → 4-6 GB of memory, an ordinary server with no GPU (Ollama / llama.cpp) · fine-tuning — LoRA on a single GPU

"Hard for a human" ≠ "hard for a model"

Data sensitivity is uncovered by asking, not assumed

The Samsung leak — and models already in your pocket

✗ Samsung, 2023

Allowed ChatGPT on 03/11/2023

By 03/30 (~19 days) — 3 leaks: 2× manufacturing code, 1× meeting transcript

Full ban on generative AI on devices (May 2023)

✓ Small models in your pocket

Apple Intelligence — ~3 billion parameters on-device (iOS 18.1, October 2024, iPhone 15 Pro+): text summarization and editing offline

Google Gemini Nano — 1.8-3.25 billion parameters (Pixel/Android): summarization in Recorder, Magic Compose — offline

Small models are already in the pocket of half this room

Your tasks



Does anyone here have a similar task from work — one where the question was whether AI was needed at all, and which kind?

It's enough to state it in two or three sentences, the way a client would actually bring it to you

What to take away

Seven lessons, one per case — nothing new, just a compilation



Vendor documents

- Two examples aren't a sample



Personal data in logs

- The regex/model boundary — inside the task



Support assistant

- An organizational constraint settles it just as well as technical ones



Product descriptions

- Being built-in beats a standalone product



Meeting minutes

- Requirements drive the architecture, not fashion



Digest

- Many sources $\neq$ RAG



Sales calls

- Hard for a human $\neq$ hard for a model

Homework reading

Optional, but there'll be something to discuss for each story



MAIN

The Klarna saga, 2023-2025

An AI support assistant: from an impressive launch to the admission "we went too far" and a return to hiring people.



NYC MyCity chatbot, 2024

A city chatbot for small businesses that gave legally incorrect advice. Investigated by The Markup.



Bonus, for the mood

A car dealership hooked up a chatbot on a third-party API with no guardrails — and the chatbot "sold" an SUV for one dollar.

If there's time left today, we'll work through part of this live

Klarna, 2023-2025: one story — four forks in the road

1 AI or not

Routine tickets — formalizable, a good candidate; but cost optimization tanked quality

2 Built-in or custom

Assistant in the existing support channel, external API (OpenAI) — not a standalone product

3 Architecture

Chat assistant with escalation to a human, not an autonomous agent

4 Inference

External cloud API with support customer data

Launched February 2024 · 30 days: 2.3 million chats, 67% of requests automated · claimed "the equivalent of 700 agents"/\$40M (2024) → "853 agents"/\$60M (Q3 2025) — denominator not disclosed

"We went too far" — CEO Sebastian Siemiatkowski, Bloomberg, May 2025

News deduplication



"Users are complaining to support that the feed is cluttered — for one event they see ten nearly identical headlines in a row from different outlets. We want to merge them into one card. There are already more than a hundred sources, all coming in through a single RSS aggregator, the headline and publication-time format is the same for all of them. We're adding a few more this month. Separately, the editor asked to update the design of the source card on the site, but that's not related to deduplication. The budget for new features this quarter is modest."



Ten nearly identical headlines → one merged event card

What technology do you need?

Deduplication — breakdown



Fuzzy matching / shingling — ~90%

Comparing texts by overlapping fragments, with no call to a model

Cheap and predictable



LLM — only the remainder

Two messages about the same event written in different words, sharing almost no fragments

Paraphrase with no lexical overlap

Cheap deterministic filter first — AI on what it didn't catch



Today you decided what to build.

Lecture 2 — how the model is built inside